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Smart Data Platform

Real-Time Data Lakehouse for Modern Analytics

A unified analytics platform that captures every change in your operational database in near real time, stores it as governed Iceberg tables, and delivers insights through SQL and self-service dashboards — without expensive proprietary warehouses or overnight batch delays.

Near-Real-Time CDC
Data queryable within minutes, not overnight batches

100% Open Source
No vendor lock-in. Deploy on-prem or any cloud

Governed Lakehouse
ACID tables, schema evolution, audit snapshots

Self-Service BI
Business dashboards via Apache Superset

Payment-Ready
Merchants, transactions, settlements, KYC, chargebacks

Deploy in 1 Week
Containerized platform, first table live end-to-end

1. What Is This?

> The **Smart Data Platform** is an enterprise-grade **data lakehouse** — a unified system that combines data lake flexibility, data warehouse reliability, and streaming speed into one open-source platform.

Data Lake Flexibility

Store all data cheaply at scale on S3-compatible object storage in open Parquet format.

Warehouse Reliability

ACID transactions, schema evolution, and versioned snapshots via Apache Iceberg.

Streaming Speed

Continuous CDC — every insert, update, delete captured and queryable within minutes.

2. The Challenge You Face Today

Traditional Analytics

- Reports reflect yesterday's data — decisions are stale
- Weeks to add a new table or metric (full ETL project)
- Expensive proprietary warehouse with rising license costs
- Engineering bottleneck for every new dashboard
- No audit trail when data changes or discrepancies appear
- Data siloed across systems with no single source of truth

Smart Data Platform

- Data fresh within minutes — real-time operational insight
- New tables onboarded in hours via dynamic CDC connectors
- 100% open-source — pay only for infrastructure
- Self-service dashboards in Superset for business users
- Full CDC history + Iceberg point-in-time snapshots
- One governed lakehouse for all teams and reports

3. How Does It Work?

Every insert, update, or delete in PostgreSQL is automatically streamed into the analytical lakehouse. No batch jobs. No manual exports. No engineering intervention per report.

PostgreSQL Source DB › **Debezium** CDC Capture › **Kafka** Event Stream › **Iceberg** Lakehouse Write › **Trino** SQL Engine › **Superset** Dashboards

Step	What Happens	Component	Latency
1	Business application creates or updates a record in the operational database	PostgreSQL 16	Immediate
2	Row-level change captured from the database Write-Ahead Log (logical replication)	Debezium CDC	Seconds
3	Change published as a durable, replayable event on a Kafka topic	Apache Kafka 4.1	Seconds
4	Event written as a versioned Parquet file; Iceberg snapshot committed to catalog	Iceberg Sink → MinIO + Nessie	Seconds–min
5	Analysts run SQL queries or business users view live dashboards	Trino 480 + Superset	On demand

4. Live Example — Merchant Onboarding

Action	System Response	Verification
App inserts new merchant into PostgreSQL	Debezium captures INSERT within seconds	Kafka UI: topic analytics.public.merchants
CDC event published to Kafka	Iceberg sink writes Parquet to MinIO, commits snapshot	MinIO Console: warehouse/bronze/merchants/
Trino query runs	New merchant row returned immediately	SELECT * FROM iceberg.bronze.merchants
Superset dashboard refreshes	Merchant count KPI updates automatically	Superset: Merchant Onboarding dashboard

5. Architecture Diagram



6. What Will Be Set Up For You

We deliver a fully containerized, production-pattern platform — 9 services orchestrated via a single Docker Compose configuration, ready to run on your infrastructure.

#	Service	Port	Role	Who Uses It
1	PostgreSQL 16	5432	Operational database; CDC source (wal_level=logical)	Applications, Debezium
2	Apache Kafka 4.1	9092	Durable event streaming backbone (KRaft mode)	Connect, Kafka UI
3	Kafka Connect	8083	Debezium CDC source + Iceberg lakehouse sink connectors	Engineering
4	Kafka UI	8080	Visual monitoring of topics, messages, connector health	Operations
5	MinIO	9000/9001	S3-compatible object storage for all Parquet data files	Iceberg, Trino
6	Project Nessie 0.108	19120	Versioned Iceberg catalog — schemas, snapshots, branches	Connect, Trino
7	Trino 480	8081	Distributed SQL — query lakehouse and source DB together	Analysts, Superset
8	Apache Superset	8088	Self-service BI dashboards and ad-hoc reports	Business users
9	MinIO Client (mc)	—	Auto-creates warehouse storage bucket on platform startup	Automated init

Connectors Configured

Debezium CDC Source Connector

PostgreSQL → Kafka. Captures every INSERT, UPDATE, DELETE via logical replication. Topics: *analytics.public.<table>*. Initial snapshot + continuous streaming.

Iceberg Sink Connector

Kafka → Lakehouse. Writes versioned Parquet files to MinIO, commits snapshots via Nessie. Explicit per-table or dynamic auto-create for all source tables.

What You Provide

Infrastructure

Docker-capable server, 8 GB+ RAM, ports 5432, 8080–8088, 9000–9001, 9092, 19120

Runtime

Docker and Docker Compose installed on the host

Source Data

PostgreSQL tables in the public schema (merchants, transactions, etc.)

7. Advantages You Will Get

Near-Real-Time Insights
Decisions based on minutes-old data, not yesterday's batch report. GMV, success rates, and settlement status always current.

Single Source of Truth
One governed Iceberg dataset for all teams, reports, and dashboards. No conflicting numbers across departments.

Lower Total Cost of Ownership
100% open-source stack. No per-query, per-seat, or proprietary warehouse licensing fees. Pay only for infrastructure.

No Vendor Lock-In
Deploy on-premises, AWS, Azure, or GCP. MinIO maps directly to S3. Same architecture everywhere.

Self-Service Analytics
Business users build and refresh dashboards in Superset without submitting engineering tickets or waiting for ETL.

Audit & Compliance Ready
Full CDC change history in Kafka. Iceberg point-in-time snapshots for regulatory reporting and dispute resolution.

Faster Time-to-Market
Add a new PostgreSQL table and the pipeline auto-follows (dynamic connector). Hours, not weeks.

Industry-Proven Architecture
Same lakehouse patterns used by Netflix, Uber, Airbnb, Stripe, and Apple. Enterprise-grade from day one.

8. Data Organization — Medallion Architecture

Bronze — Raw Landing
As-is CDC replication from source. Full insert/update/delete history. Auto-populated by the pipeline — zero manual ETL.
merchants, transactions, settlements, chargebacks, kyc_documents

Silver — Cleansed & Conformed
Deduplicated, validated business entities. Standardized schemas. Quality rules applied. Ready for analytics.
payments, merchant_risk, clean_settlements

Gold — Business Metrics
Executive KPIs and aggregates. Regulatory reporting tables. Optimized for dashboard performance.
daily_gmv, payment_success_rate, settlement_summary

9. Use Cases for Your Payment Platform

Business Use Case	Source Tables	Outcome / Dashboard	Layer
Merchant onboarding funnel	merchants, kyc_documents	Live onboarding completion rate by stage	Bronze → Gold
Payment success monitoring	transactions	Real-time success/failure rate by gateway and merchant	Bronze → Gold
Settlement reconciliation	settlements, transactions	T+1 settlement accuracy and exception alerts	Silver → Gold
Chargeback analysis	chargebacks, merchants	Chargeback ratio alerts by merchant and category	Silver → Gold
Executive KPI dashboard	Gold layer aggregates	Daily GMV, active merchants, revenue trends	Gold
Regulatory & audit reporting	All domains	Point-in-time data snapshots for compliance	All layers
Fraud & risk scoring feed	transactions, merchant_risk	Governed datasets for ML model training	Silver

10. Technology Stack — Enterprise Credibility

Category	Technology	Version	Adopted By	License
Database	PostgreSQL	16	Apple, Instagram, Spotify	Open
Event Streaming	Apache Kafka	4.1	LinkedIn, Netflix, Uber	Apache 2.0
CDC	Debezium	2.7	Red Hat, Zalando, Airbnb	Apache 2.0
Table Format	Apache Iceberg	1.x	Apple, Netflix, Adobe	Apache 2.0
Data Catalog	Project Nessie	0.108	Apache Iceberg ecosystem	Apache 2.0
Object Storage	MinIO	Latest	Siemens, Adobe (S3-compatible)	AGPL v3
SQL Engine	Trino	480	Meta, Twitter, Comcast	Apache 2.0
Business Intelligence	Apache Superset	Latest	Airbnb, Lyft, Dropbox	Apache 2.0

11. Implementation Roadmap

1. **Foundation — Week 1**
   Deploy all 9 services via Docker Compose. Configure Debezium CDC and Iceberg sink connectors. Verify first table (merchants) flowing end-to-end: PostgreSQL → Kafka → Iceberg → Trino → Superset.
2. **Core Domains — Weeks 2–3**
   Onboard transactions, settlements, chargebacks, and KYC tables. Complete Bronze layer. Build operational dashboards in Superset for payments and merchant monitoring.
3. **Analytics Maturity — Weeks 4–6**
   Build Silver (cleansed entities) and Gold (KPI aggregates) layers via Trino SQL. Deliver executive dashboards: daily GMV, payment success rate, settlement summary, merchant onboarding funnel.
4. **Production Hardening — Weeks 7–10**
   Enable TLS, SSO/LDAP, secrets vault. Migrate storage to cloud S3. Deploy on Kubernetes with high availability. Implement Prometheus + Grafana monitoring and alerting.

Phase	Duration	Key Deliverable	Success Criteria
1 — Foundation	~1 week	Platform live; first table end-to-end	INSERT in Postgres appears in Superset dashboard
2 — Core Domains	~2 weeks	5 domain tables in Bronze; ops dashboards	All payment tables streaming; 3+ dashboards live
3 — Analytics Maturity	~3 weeks	Silver/Gold layers; executive KPIs	Leadership dashboard with daily GMV and success rate
4 — Production	~4 weeks	Security, cloud, HA, monitoring	Pen-test passed; 99.9% uptime SLA met

12. Security & Governance

Built-In Governance

- Immutable CDC event log — every change recorded in Kafka
- Iceberg table versioning — query data at any historical point
- Nessie catalog branches — isolate dev, test, and production data
- Schema evolution tracking — audit all structural changes
- Role-based access via Trino and Superset (production)

Production Security Roadmap

Area	Enhancement
Encryption	TLS on Kafka, Trino, Nessie, Connect
Authentication	SSO / LDAP for Trino and Superset
Secrets	HashiCorp Vault or cloud secrets manager
Network	Private subnets; restricted inter-service traffic
Audit	Centralized logging for all data access events

Scalability — POC to Production

Component	POC (Day 1)	Production
Kafka	1 node (KRaft)	Multi-broker cluster, RF ≥ 3
Object Storage	MinIO (single node)	AWS S3 / Azure Data Lake / GCS
Catalog	Nessie (in-memory)	Nessie with persistent JDBC backend
SQL Engine	1 Trino coordinator	Trino cluster with dedicated workers
Connect	1 Connect worker	Horizontally scaled Connect cluster
Orchestration	Docker Compose	Kubernetes with auto-scaling
Monitoring	Kafka UI	Prometheus + Grafana + PagerDuty

Deliverables You Receive

- Running 9-service platform (Docker Compose)
- CDC + Iceberg connector configurations (ready to register)
- Trino catalogs: iceberg (lakehouse) + postgresql (source)
- Superset with Trino integration pre-configured
- Kafka UI for pipeline observability
- Architecture documentation and setup runbook
- Bronze / Silver / Gold schema structure in Trino

13. Before vs After — Summary

Dimension	Before (Traditional)	After (Smart Data Platform)	Business Impact
Data freshness	Hours to 24+ hours	Minutes	Faster decisions, reduced risk exposure
Add new data source	Days to weeks	Hours (add table + auto-connector)	Accelerated product analytics
Analytics cost	High license + infra	Open-source infra only	60–80% cost reduction potential
Vendor dependency	High (proprietary DW)	None (portable to any cloud)	Negotiating leverage; no lock-in
Audit capability	Limited / manual	Full CDC + Iceberg snapshots	Regulatory compliance confidence
Self-service BI	Engineering bottleneck	Native Superset dashboards	Business team autonomy
Schema changes	Break pipelines	Iceberg schema evolution	Zero-downtime schema updates
Disaster recovery	Complex backup/restore	Kafka replay + immutable storage	Faster recovery; no data loss

14. Access URLs After Deployment

Tool	URL	Primary Users	Purpose
Apache Superset	http://localhost:8088	Business users, leadership	Dashboards, charts, ad-hoc reports
Trino SQL	http://localhost:8081	Data analysts, engineers	SQL queries over lakehouse and source DB
Kafka UI	http://localhost:8080	Operations, engineering	Monitor topics, connectors, pipeline health
MinIO Console	http://localhost:9001	Engineering	Inspect Parquet files and storage usage
Kafka Connect API	http://localhost:8083	Engineering	Register and manage CDC/sink connectors
PostgreSQL	localhost:5432	Applications	Operational database (source of truth)

15. Next Steps

#	Action	Owner	Target
1	Review this document with business and technical stakeholders	Joint session	Week 0
2	Confirm priority data domains and source tables (merchants, transactions, etc.)	Client business team	Week 0
3	Provision Docker host (8 GB+ RAM, required ports open)	Client infrastructure	Week 0
4	Deploy platform and register CDC + Iceberg connectors	Delivery team	Week 1
5	Joint end-to-end verification — live merchant insert → dashboard	Joint session	Week 1
6	Onboard core payment domain tables (Phase 2)	Delivery team	Weeks 2–3
7	Build executive KPI dashboards in Superset (Phase 3)	Delivery + client analysts	Weeks 4–6
8	Plan production hardening — security, cloud, HA (Phase 4)	Joint planning	Week 7+

Ready to transform your analytics?

The Smart Data Platform can be deployed and verified with your first live data table within one week.
Contact the delivery team to schedule a joint kick-off session.

Glossary

Term	Definition
CDC	Change Data Capture — detects and streams row-level database changes in real time
Lakehouse	Architecture combining data lake storage flexibility with data warehouse governance
Apache Iceberg	Open table format providing ACID transactions, schema evolution, and time travel
Project Nessie	Git-like catalog for Iceberg tables with branch, tag, and snapshot semantics
Medallion Architecture	Data layering: Bronze (raw) → Silver (cleansed) → Gold (business metrics)
Trino	High-performance distributed SQL query engine for lakehouse and federated queries